



Experimental demonstration of indoor 3D optical wireless positioning and azimuth estimation using steerable line lasers

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ABSTRACT

We present an indoor three-dimensional (3D) optical wireless positioning and azimuth estimation (OWPAE) scheme that utilizes steerable line lasers. This approach enables the simultaneous acquisition of precise 3D coordinate and azimuth of a mobile receiver without requiring a priori knowledge of the radiation pattern of the light source and photo-detector (PD) characteristics. Our work begins with an analysis of the feasibility of using line lasers for OWPAE. Subsequently, by leveraging the geometric relationship between two line lasers and a pair of PDs, we establish the OWPAE model, considering receiver tilt, and design the frame structure for conveying essential information. To evaluate the effectiveness of our proposed OWPAE scheme, we set up an experimental platform. Experimental results confirm that when the receiver is oriented upwards, the system accurately determines the receiver's location with a mean 3D positioning error of 6.3 cm and its azimuth with a mean azimuthal error of 2.8°. When the receiver tilts at a polar angle of 30°, the system remains effective, yielding a mean 3D positioning error of 8.3 cm and a mean azimuthal error of 5.6°.

1. Introduction

With the rapid evolution of Mobile Internet and Industrial Internet-of-Things, the demand for high-accuracy indoor positioning systems (IPSS) has surged in the past decade [1]. These systems play a crucial role in providing precise location-based services for various applications, ranging from centimeter-level indoor navigation and virtual reality entertainment in daily life to applications in industrial manufacturing, such as automatic guided vehicles and precise robot control in smart factories. Traditional global satellite navigation systems face challenges in delivering accurate positioning in indoor or underground environments due to signal attenuation [2]. While radio frequency (RF) technologies like Wi-Fi, Bluetooth, and ZigBee have been extensively researched for IPSS, their positioning accuracy is often limited to the sub-meter level [3]. This limitation arises from inherent challenges in RF channels, including multipath propagation and time-variable properties. In recent years, the emergence of light-emitting diode (LED) indoor lighting and the rapid advancements in optical wireless communication (OWC) technology have propelled optical wireless positioning (OWP) as a promising solution for IPSS

[4–6]. OWP offers advantages, including precise positioning, cost-effectiveness, and system integration [7–11]. This enhanced accuracy is associated with the characteristics of optical wireless channels, featuring line-of-sight (LOS) propagation and time-invariant properties. These features enhance the stability of received optical signals in light-based IPSS, resulting in superior accuracy compared to RF alternatives [10].

Conventional optical wireless positioning (OWP) systems commonly utilize trilateration or triangulation for achieving centimeter-level two-dimensional (2D) or three-dimensional (3D) positioning. Ranging methods, such as time of arrival (TOA), time difference of arrival (TDOA), received signal strength (RSS), and angle of arrival (AOA), present available options [12–14]. Despite theoretical support for TOA and TDOA [15–17], these methods are rarely implemented in practical systems due to challenges in achieving perfect synchronization between transceivers. RSS-based OWP systems [18–23] rely on a priori knowledge of the LED radiation pattern and photo-detector (PD) characteristics, using received signal power to estimate the distance between transceivers and employing trilateration for positioning. Similarly, in AOA-based OWP systems, the estimation of incidence angles of light

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signals relies on the LED radiation pattern and PD characteristics. These incidence angles are then utilized to determine the unique position of the receiver [24,25]. However, current RSS and AOA based approaches often assume an ideal optical wireless channel model, assuming a perfect LED radiation pattern (often Lambertian) and fixed PD characteristics for taking RSS or AOA measurements [18–25]. In practice, commercial LEDs usually do not follow the perfect Lambertian radiation pattern. Different OWP systems also use LEDs and PDs produced by various manufacturers, necessitating the calibration of parameters in the optical wireless channel model before deploying OWP systems [26,27]. Calibration, especially of the LED radiation pattern, is crucial to prevent degradation of positioning accuracy. However, this calibration process poses challenges for deploying OWP in different indoor scenarios, limiting system flexibility. Additionally, conventional OWP systems typically require a minimum of three LED light sources for trilateration or triangulation, further restricting system flexibility when the number of LEDs is insufficient. To enhance system flexibility, new approaches need to be proposed.

Although 2D positioning is more fundamental and widely used, with the receiver height usually known, in practice, the receiver often moves randomly in 3D space, and its height information is frequently unknown and needs to be obtained. Therefore, increasing attention has been directed toward 3D OWP [28–36]. To extend OWP from 2D to 3D, one type of approach involves increasing the number of PDs and combining them into a dedicated designed array. This allows for the reception of more information from higher dimensions to estimate the 3D coordinates of the receiver. For example, in Refs. [28,29], 3D OWP schemes using AOA and RSS were proposed, where the receivers consist of an array composed of three and four tilted PDs, respectively. In Ref. [30], a 3D OWP scheme based on beacon LED coordinates estimator was proposed, utilizing four tilted PDs to form each PD array, with at least two arrays selected to detect the AOA. In these schemes, the receiver requires at least three tilted PDs to form the PD arrays, leading to an increase in system complexity. Moreover, the orientation of each PD in the arrays should be known in advance to facilitate OWP algorithms, requiring additional calibration or measurement steps. Another type of approach is to use additional equipment to assist 3D OWP. For example, in Ref. [31], to achieve 3D positioning, three lighthouses are installed in the X-, Y-, and Z-axes, each comprising two steerable lasers and a rotating platform. These lasers scan in 2D with a rotating mirror and a rotating platform, and the PD estimates the 3D coordinates based on the duration of received signals. In Ref. [32], the proposed 3D OWP scheme involves not only using LEDs to transmit positioning information, with the PD to capture RSS, but also incorporating an additional steerable laser steered in 2D to assist in estimating height information. In Refs. [33,34], acceleration sensors were employed to obtain the azimuthal angle of mobile terminals. This information is then combined with the RSS or AOA received by the PD, allowing for trilateration or triangulation to achieve 3D OWP. However, the use of additional equipment increases both the cost and complexity of the system. There were also 2D or 3D OWP schemes using image sensors as receivers [35–38]. Compared with image sensors, PDs have higher response speed and offer stronger privacy. Therefore, this paper focuses on a 3D OWP system based on PDs. Besides positioning, azimuthal information of user devices has recently gained increasing importance for location-based applications, such as indoor navigation and automatic parking. Therefore, accurate azimuth estimation of the receiver is crucial. Joint receiver location and orientation estimation schemes have been proposed for OWP systems [39,40]. However, these schemes require a large number of LED light sources, such as 20 LEDs in Ref. [39] and 48 LEDs in Ref. [40]. Furthermore, they were designed to support only 2D OWP. Consequently, there are still very few OWP systems capable of supporting simultaneous 3D positioning and azimuth estimation with a simple system architecture.

To overcome these challenges and improve system flexibility, this paper introduces an indoor 3D OWP and azimuth estimation (OWPAE)

scheme. This approach employs only two line lasers as transmitting antennas and a pair of PDs for signal reception. The steering angles of the line lasers are controlled by two one-dimensional (1D) steering devices, respectively. By measuring the steering angles of the line lasers when PDs receive the emitted signals, accurate estimation of both the 3D coordinates and the azimuthal angle of the receiver can be achieved. In comparison to traditional OWP schemes, our proposed scheme not only reduces the number of transceivers required but also separates OWP from LED lighting facilities, decreasing its reliance on the optical wireless channel model. This enables the independent deployment of an OWP system without the need to measure or fit the LED radiation pattern. As indicated in Table 1, our scheme is the only one supporting simultaneous 3D OWP and azimuth estimation without a priori knowledge of the radiation pattern of the light source.

The original concept of utilizing steerable line lasers for achieving 3D OWPAE was initially introduced in our conference paper [41], and its feasibility was confirmed through numerical simulations. In this work, we elaborate on this concept and develop the OWPAE model, considering the scenario of a tilted receiver. Furthermore, an experimental setup is implemented to verify the feasibility of using line lasers for OWPAE and to validate the scheme in real-world scenarios. We investigate 1/8 of the receiving plane in an indoor space with dimensions of $4\text{ m} \times 4\text{ m} \times 2.8\text{ m}$, analyzing the influence of receiver height, the distance between PDs, and steering angle resolution on the positioning and azimuth estimation accuracy of the system. Results demonstrate that this scheme effectively achieves accurate 3D OWPAE. Specifically, when the receiver is facing upwards, the scheme achieves a mean 3D positioning error of 6.3 cm and a mean azimuthal error of 2.8° . When the receiver tilt angle is 30° , the scheme attains a mean 3D positioning error of 8.3 cm and a mean azimuthal error of 5.6° . Results also indicate that increasing the receiver height, the distance between PDs, and the steering angle resolution improves the accuracy of the OWPAE system.

The rest of this paper is organized as follows. Section 2 analyzes the feasibility of adopting line lasers for OWPAE. Section 3 describes our proposed OWPAE scheme in detail. Section 4 introduces the experimental setup and presents the experimental results and discussions. Finally, Section 5 concludes the paper.

2. Analysis of the feasibility of adopting line lasers for OWPAE

A line laser emits a narrow and straight laser beam in the form of a line, unlike a point or diffuse beam. This line-shaped laser beam is commonly used in various applications for tasks like alignment, measurement, and reference purposes, making it a crucial tool in industrial, construction, and engineering applications. In this paper, we utilize two line lasers as transmitters for conveying positioning information in our OWPAE system, with a focus on accurately recovering positioning information at the receiver. However, the nature of the line-shaped beam emitted by line lasers can lead to the distribution of received power at the receiver across the light spot. This distribution may result in a relatively low received voltage at the receiver, potentially affecting the accuracy of positioning and azimuth estimation, especially when the terminal moves within the receiving plane, particularly at a significant distance from the line laser. In such cases, the receiver may struggle to demodulate the positioning information, leading to the breakdown of the OWPAE functions.

To assess the feasibility of using line lasers for OWPAE within a given receiving plane, it is crucial to establish the relationship between the driving voltage and the received voltage of a PD by measuring and fitting the beam propagation characteristics of line lasers. This model helps understand the connection between the received voltage of a PD and the driving voltage of the line laser, determining whether the receiver can successfully capture and demodulate the positioning information from the emitted signal of line lasers for a given driving voltage. Only by ensuring that the positioning information sent by each line laser can be successfully received and recovered can we construct

Table 1

Comparison between different OWP schemes based on PDs.

Reference	Transmitter	Receiver	Rely on the radiation pattern of the transmitter	2D or 3D OWP	Support azimuth estimation	Support receiver tilt
[19]	3 LEDs	1 PD	Yes	2D	No	No
[20]	4 LEDs	1 PD	Yes	2D	No	No
[21]	2 LEDs + 1 mirror	1 PD	Yes	2D	No	No
[22]	3 LEDs	1 PD	Yes	2D	No	No
[23]	4 LEDs	2 PDs	Yes	2D	No	No
[24]	4 LEDs	1 PD	Yes	2D	No	Yes
[25]	11 LEDs	4 PDs	Yes	3D	No	Yes
[28]	1 LED array	3 PDs	Yes	3D	No	Yes
[29]	4 LEDs	4 PDs	Yes	3D	No	Yes
[30]	1 LED	8 PDs	Yes	3D	No	Yes
[31]	6 lasers + 3 steering devices (in 2D)	1 PD	No	3D	No	N/A
[32]	4 LEDs + 1 laser + 1 steering device (in 2D)	1 PD	Yes	3D	No	No
[33]	37 LEDs + 1 accelerometer	1 PD	Yes	3D	No	No
[34]	3 LEDs	1 PD + 1 accelerometer	Yes	3D	No	Yes
[39]	20 LEDs	1 PD	Yes	2D	Yes	No
[40]	48 LEDs	6 PDs	Yes	2D	Yes	Yes
This work	2 line lasers + 2 steering devices (in 1D)	2 PDs	No	3D	Yes	Yes

the OWPAE system using two line lasers, as detailed in Section 3.

As shown in Fig. 1(a), we consider a scenario where the line laser is treated as a point source, and a PD is used to detect the optical signal emitted by the line laser. In this setup, we assume that the vertical distance between the line laser and the receiving plane is h , and the laser emits light with a fan angle of 2ψ . When we draw a perpendicular line from the line laser to the receiving plane and denote the angle between this perpendicular line and the direction of laser emission as γ , the beam emitted by the line laser, propagating in free space, forms a line-shaped beam on the receiving plane with a length of $2h\tan\psi/\cos\gamma$. Ideally, the received power is uniformly distributed along this beam, allowing us to express the average electrical power received by the PD as.

$$P_{ave}(V_t) = \frac{(\eta\xi V_t I_t(V_t))^2 R d_T}{2h \tan \psi / \cos \gamma}, \quad (1)$$

where d_T is the aperture size of the PD, η is the responsivity of the PD at the wavelength used by the line laser, ξ is the electro-optic conversion efficiency of the laser, V_t is the driving voltage of the laser, I_t is the driving current of the laser, which can be seen as a function of V_t , and R is the load resistance value of the PD. However, in practical scenarios, the distribution of received electrical power will gradually decrease from the maximum value $P_{max}(V_t)$ at the center of the beam towards both sides. Let us assume $P_{max}(V_t) = \mu P_{ave}(V_t)$, where the ratio μ can be adjusted to different values based on various models of line lasers. Consequently, we can compute the maximum received voltage $V_{max}(V_t)$ corresponding to $P_{max}(V_t)$ as follows:

$$V_{max}(V_t) = M \sqrt{R P_{max}(V_t)} = M \sqrt{R \mu P_{ave}(V_t)}, \quad (2)$$

where M is the amplification of the received voltage at the PD. It is important to note that as the received power diminishes along both sides of the beam, the received voltage similarly decreases and follows a similar distribution trend. Thus, by taking inspiration from the Gaussian beam model mentioned in Ref. [42], we can establish a relationship model between the received voltage and the driving voltage as follows:

$$V_r(V_t, x) = V_{max}(V_t) \exp\left(-\frac{x^2}{h \tan \psi / \cos \gamma}\right) \\ = MR \eta \xi V_t I_t(V_t) \sqrt{\frac{\mu d_T}{2h \tan \psi / \cos \gamma}} \exp\left(-\frac{x^2}{h \tan \psi / \cos \gamma}\right), \quad (3)$$

where x is the distance between the PD and the center of the line-shaped beam. Based on Eq. (3), to calculate the received voltage at the PD, it is essential to have information about the voltage-current characteristic of the laser and the electro-optic conversion efficiency. Therefore, we conduct measurements by varying the driving voltage to determine the driving current associated with the line laser. Subsequently, we establish a voltage-current characteristic curve, as illustrated in Fig. 1(b). The values of the parameters used in our measurements in Fig. 1 can be found in Table 2. As observed in Fig. 1(b), the driving current increases as the driving voltage gradually rises. The linear working area is evident when the driving voltage is from 2.06 V to 5.86 V. We utilize this voltage-current characteristic to calculate the electro-optic conversion efficiency of the line laser at various driving voltages. By applying the min-max normalization method and considering the electro-optic conversion efficiency as 1 when the driving voltage is 2.06 V, we fit a normalized electro-optic conversion efficiency curve, as depicted in Fig. 1(b). This curve illustrates that the electro-optic conversion efficiency gradually decreases as the driving voltage increases. This decrease is attributed to the rise in heat generated by the laser as the driving voltage increases, leading to a higher proportion of electrical energy being converted to heat and, consequently, a reduction in electro-optic conversion efficiency.

To validate the accuracy of Eq. (3), we conduct measurements of the received voltage at various positions within the laser beam. As shown in Fig. 1(a), we select four representative test points. When γ is 0° , we select point A at the beam's center and point B, located 2 m away from A. When γ is 45° , we select point A' at the beam's center and point B', which is 2 m away from A'. Fig. 1(c) displays a comparison between the theoretical and experimental values of received voltages at the four test locations under varying driving voltages. The results indicate a close agreement between the experimental and theoretical values. For instance, when the drive voltage is 4.5 V, the theoretical values of the received voltage at points A, B, A', and B' are 3.02 V, 0.74 V, 2.53 V, and 0.94 V, respectively, closely matching the actual measured values of

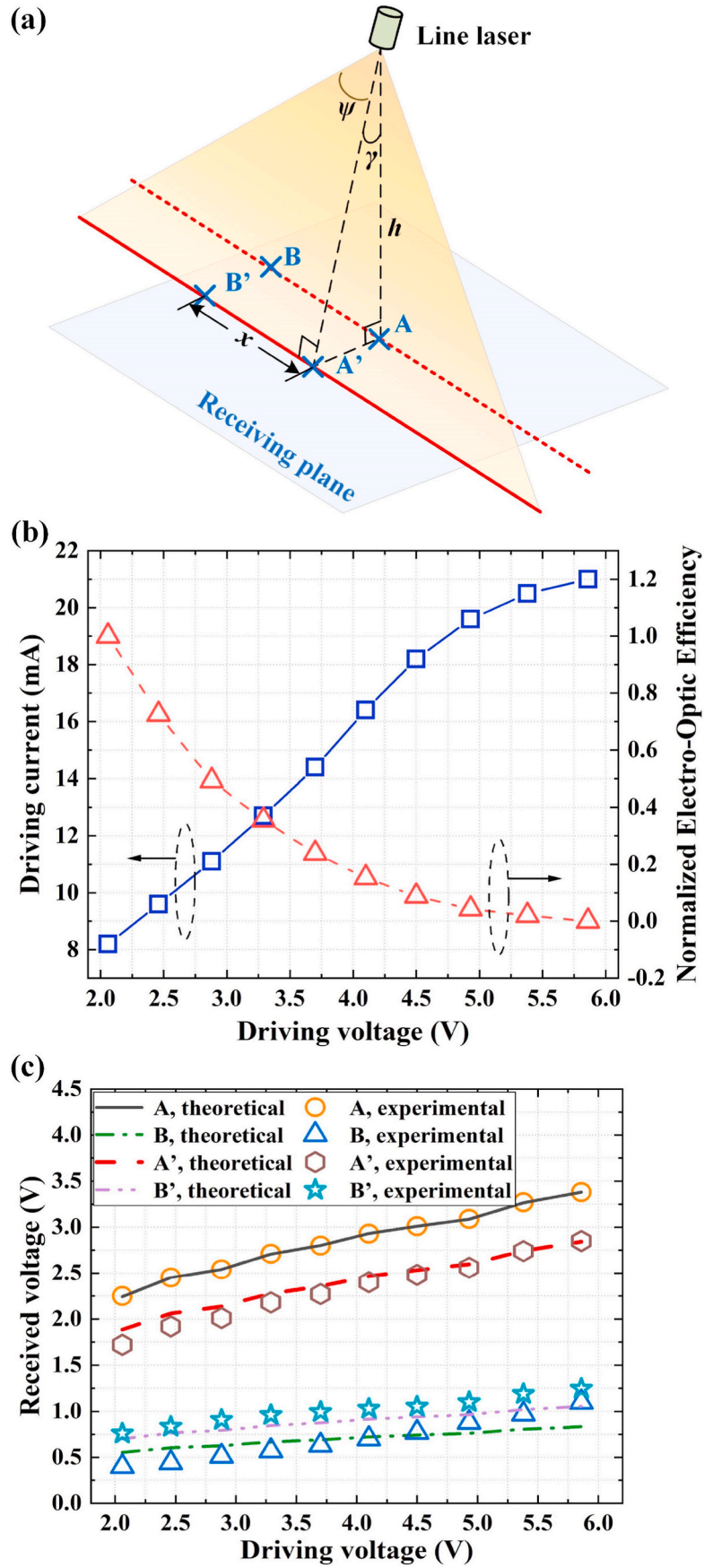


Fig. 1. (a) Experimental setup to analyze the feasibility of adopting line lasers for OWPAE; (b) the voltage-current characteristic and the electro-optic conversion efficiency of the line laser; (c) the theoretical and experimental values of received voltages at the four test locations under different driving voltages.

Table 2

Parameters for analyzing the feasibility of adopting line lasers for OWPAE.

Parameters	Values
Responsivity of the PD (η)	0.45 A/W
Aperture size of the PD (d_p)	9.8 mm
Load resistance value of the PD (R)	50 Ω
Voltage amplification at the PD (M)	10^3
Half of fan angle of the line laser (ψ)	55°
The ratio of maximum received electrical power to average received electrical power (μ)	1.2
Vertical distance between the laser and the receiving plane (h)	2 m

3.01 V, 0.77 V, 2.48 V, and 1.05 V. This confirms the validity of our proposed model in Eq. (3). Therefore, by employing Eq. (3), we can assess whether the received voltage aligns with the positioning requirements when the receiver is placed at different locations in a $4 \text{ m} \times 4 \text{ m}$ receiving plane. As an example, when the PD is positioned at $x = 2 \text{ m}$ and $\gamma = 45^\circ$, as at point B', it successfully captures the transmitted signal.

This implies that when the PD is located within the range of $0 \text{ m} \leq x \leq 2 \text{ m}$ and $0^\circ \leq \gamma \leq 45^\circ$, it can consistently receive the transmitted signal. This specified region corresponds to the $4 \text{ m} \times 4 \text{ m}$ receiving plane considered in Section 4, constituting the primary area of interest in this paper.

3. Proposed OWPAE scheme

Now, let us introduce the proposed OWPAE scheme based on line lasers. Fig. 2(a) depicts the architecture of our OWPAE system, comprising a transmitter and a receiver. The transmitter is equipped with two steerable line lasers, TX1 and TX2, mounted on the ceiling. The receiver is mobile within the receiving plane, and a pair of PDs, PD1 and PD2, are positioned at both ends. At the transmitter, TX1 is located at the coordinate (x_{t1}, y_{t1}, z_{t1}) and can be steerable along the X-axis, while TX2 is located at the coordinate (x_{t2}, y_{t2}, z_{t2}) and can be steerable along the Y-axis. Both line lasers emit line-shaped beams with wide fan angles and alternately scan the entire receiving plane rapidly. Throughout this scanning process, quantized values of their dynamic steering angles, α

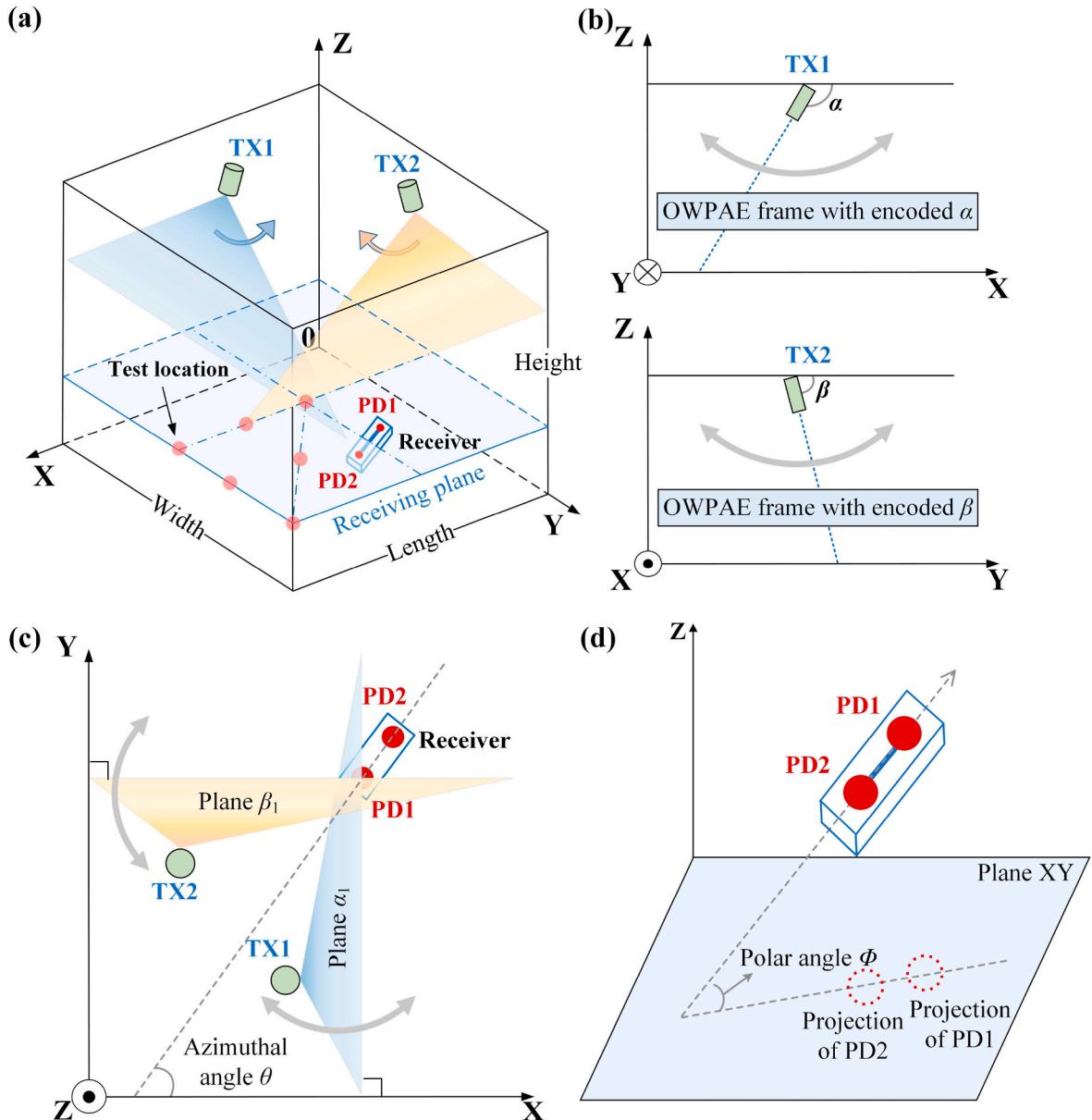


Fig. 2. (a) Proposed OWPAE system architecture; (b) side view; (c) top-down view; and (d) description of the polar angle for a tilted receiver.

and β , are encoded within the OWPAE data frames and emitted by the line lasers, as shown in Fig. 2(b). This approach eliminates signal interference between TX1 and TX2, ensuring that the values of α and β are accurately received. Additionally, since laser transmitters can be modulated at high speeds, we can insert communication time slots alongside positioning time slots to facilitate indoor data broadcasting. As a result, line-shaped light spots carrying signals from TX1 and TX2 alternately and orthogonally project onto the receiving plane, as shown in Fig. 2(c).

At the receiver, we employ PD1 and PD2 to detect light signals. PD1 is located at the coordinate (x_{r1}, y_{r1}, z_{r1}) , and PD2 is located at the coordinate (x_{r2}, y_{r2}, z_{r2}) . The distance between them is fixed at l . The midpoint between PD1 and PD2 serves as the real position of the receiver to be predicted. The azimuthal angle of the receiver is defined as the anticlockwise angular rotation θ ($-180^\circ < \theta \leq 180^\circ$) with respect to the X-axis, with the direction pointing from the projection of PD2 towards the projection of PD1 in the plane XY, as shown in Fig. 2(c). We also assume that the receiver has the capability to tilt freely, with its polar angle defined by Φ ($0^\circ \leq \Phi \leq 90^\circ$), as illustrated in Fig. 2(d). Furthermore, we represent the receiver as an abstract line segment. By successfully estimating the 3D coordinates of both PD1 and PD2, we can calculate the position and azimuth of the receiver. As shown in Fig. 2(c), PD1 can receive OWPAE frames containing encoded angular information α_1 and β_1 emitted from TX1 and TX2, while PD2 can receive encoded α_2 and β_2 from TX1 and TX2, respectively. To quantify these angles, we allocate N_d bits for each line laser. Consequently, both the angular increments $\Delta\alpha$ and $\Delta\beta$ for steering angles are set at $180^\circ/2^{N_d}$. To illustrate this, let us consider the quantification of α_1 . Assuming $k = 0, 1, \dots, 2^{N_d}$, if $k\Delta\alpha < \alpha_1 \leq (k+1)\Delta\alpha$, then $\alpha_1 = (k+0.5)\Delta\alpha$. The same quantization process applies to α_2, β_1 , and β_2 .

When PD1 receives the encoded steering angle α_1 from TX1, it implies that PD1 is situated in a specific plane, which we refer to as plane α_1 . To define this plane, we can start by taking the intersection line of plane α_1 with the XY plane. On plane α_1 , if we draw a perpendicular line with respect to the intersection line through TX1, the coordinate of the foot of the perpendicular should be $(x_{t1} + z_{t1}\cot\alpha_1, y_{t1}, 0)$. Furthermore, the projection of the foot of the perpendicular onto the X-axis can be found at $(x_{t1} + z_{t1}\cot\alpha_1, 0, 0)$. Thus, plane α_1 is determined by three key points: TX1 at (x_{t1}, y_{t1}, z_{t1}) , the foot of the perpendicular at $(x_{t1} + z_{t1}\cot\alpha_1, y_{t1}, 0)$, and the projection point of the foot of the perpendicular at $(x_{t1} + z_{t1}\cot\alpha_1, 0, 0)$. Therefore, plane α_1 can be represented by:

$$\left(\frac{-1}{x_{t1} + z_{t1}\cot\alpha_1}\right)x + \left(\frac{-\cot\alpha_1}{x_{t1} + z_{t1}\cot\alpha_1}\right)z + 1 = 0. \quad (4)$$

Similarly, when PD1 receives β_1 from TX2, it should be situated in plane β_1 , which could be described as:

$$\left(\frac{-1}{y_{t2} + z_{t2}\cot\beta_1}\right)y + \left(\frac{-\cot\beta_1}{y_{t2} + z_{t2}\cot\beta_1}\right)z + 1 = 0. \quad (5)$$

Consequently, PD1 lies on the intersection line of planes α_1 and β_1 , and its coordinate can be expressed in the point-slope form as:

$$\frac{\hat{x}_{r1} - (x_{t1} + z_{t1}\cot\alpha_1)}{-\cot\alpha_1} = \frac{\hat{y}_{r1} - (y_{t2} + z_{t2}\cot\beta_1)}{-\cot\beta_1} = \hat{z}_{r1}. \quad (6)$$

Similarly, PD2 also lies on the intersection line of planes α_2 and β_2 , and its coordinate can be expressed as:

$$\frac{\hat{x}_{r2} - (x_{t1} + z_{t1}\cot\alpha_2)}{-\cot\alpha_2} = \frac{\hat{y}_{r2} - (y_{t2} + z_{t2}\cot\beta_2)}{-\cot\beta_2} = \hat{z}_{r2}. \quad (7)$$

Due to the fixed distance l between PD1 and PD2, we have the following constraint condition:

$$(\hat{x}_{r1} - \hat{x}_{r2})^2 + (\hat{y}_{r1} - \hat{y}_{r2})^2 + (\hat{z}_{r1} - \hat{z}_{r2})^2 = l^2. \quad (8)$$

Assuming that the values of l and Φ are already known at the receiver, we can express $\hat{z}_{r2}$ as z , and $\hat{z}_{r1}$ as $z + l\sin\Phi$, both of which are

to be estimated. Then, based on Eqs. (6)–(8), we obtain the following relationship:

$$(A^2 + C^2)z^2 + 2(AB + CD)z + B^2 + D^2 - l^2 \cos^2 \Phi = 0, \quad (9)$$

where $A = \cot\alpha_2 - \cot\alpha_1$, $B = -Az_{t1} - l\sin\Phi\cot\alpha_1$, $C = \cot\beta_2 - \cot\beta_1$, and $D = -Cz_{t2} - l\sin\Phi\cot\beta_1$. Solving for z in Eq. (9) will yield two potential solutions:

$$z_{1,2} = \frac{-(AB + CD) \pm \sqrt{2ABCD - B^2C^2 - A^2D^2 + (A^2 + C^2)l^2 \cos^2 \Phi}}{A^2 + C^2}. \quad (10)$$

Accordingly, by substituting z_1 and z_2 into Eqs. (6) and (7), we can obtain two pairs of coordinates for PD1 and PD2.

To determine the real locations for PD1 and PD2, we first rule out the solution that is outside the considered room range. Next, we compare the received angular information emitted from TX1 and TX2 to make a judgment as follows. Specifically, if $\alpha_1 \leq \alpha_2$ and $\beta_1 \leq \beta_2$, then we select the PD pair with coordinates $\hat{x}_{r1} \geq \hat{x}_{r2}$ and $\hat{y}_{r1} \geq \hat{y}_{r2}$; if $\alpha_1 > \alpha_2$ and $\beta_1 \leq \beta_2$, then we select the PD pair with coordinates $\hat{x}_{r1} < \hat{x}_{r2}$ and $\hat{y}_{r1} \geq \hat{y}_{r2}$; if $\alpha_1 \leq \alpha_2$ and $\beta_1 > \beta_2$, then we select the PD pair with coordinates $\hat{x}_{r1} \geq \hat{x}_{r2}$ and $\hat{y}_{r1} < \hat{y}_{r2}$; otherwise, if $\alpha_1 > \alpha_2$ and $\beta_1 > \beta_2$, we select the pair of PDs with their coordinates $\hat{x}_{r1} < \hat{x}_{r2}$ and $\hat{y}_{r1} < \hat{y}_{r2}$. Finally, we calculate the receiver's location as the midpoint between PD1 and PD2: $((\hat{x}_{r1} + \hat{x}_{r2})/2, (\hat{y}_{r1} + \hat{y}_{r2})/2, (\hat{z}_{r1} + \hat{z}_{r2})/2)$, and the receiver's azimuthal angle is determined as:

$$\hat{\theta} = \begin{cases} \arccos \left[\frac{\hat{x}_{r2} - \hat{x}_{r1}}{\sqrt{(\hat{x}_{r2} - \hat{x}_{r1})^2 + (\hat{y}_{r2} - \hat{y}_{r1})^2}} \right], & \hat{y}_{r1} \geq \hat{y}_{r2} \\ 360^\circ - \arccos \left[\frac{\hat{x}_{r2} - \hat{x}_{r1}}{\sqrt{(\hat{x}_{r2} - \hat{x}_{r1})^2 + (\hat{y}_{r2} - \hat{y}_{r1})^2}} \right], & \hat{y}_{r1} < \hat{y}_{r2} \end{cases}. \quad (11)$$

To ensure the reliable decoding of information at the receiver, we design a frame structure for the OWPAE scheme, as depicted in Fig. 3. This frame structure comprises five essential components: the frame header, the transmitter index, the steering angle resolution, the encoded steering angle, and the transmitter's coordinate. To prevent synchronization errors, we employ the frame header "1110010" for triple synchronization. While it is common in many OWP systems to repeat the header once to reduce positioning latency, the OWPAE frame structure in our design uses "000" to "111" to represent resolutions 9 to 16. Repeating the header only once or twice might result in the frame header being identical to the other part of the frame, leading to potential synchronization errors. For example, with a resolution of 10 and a steering angle between 20.04° and 20.21° , the encoded steering angle is "0000000001110010", which includes "1110010". Similarly, with a resolution of 15 and a steering angle between 130.078° and 130.429° , the first 14 bits of the transmitter index, the steering angle resolution, the encoded steering angle for TX2 are exactly "11100101110010". Consequently, we repeat the header three times to ensure the header's length exceeds that of the total length of the transmitter index, the steering angle resolution, and the encoded steering angle, thus preventing synchronization errors.

The decoding of the data frame is only initiated when all three synchronization attempts are successful. If not, the receiver discards the signal and awaits the next cycle. The transmitter index and the steering angle resolution are determined using a lookup table, as shown in Table 3. We distinguish between the lasers scanning along the X-axis with an identification number of "0" and scanning along the Y-axis with an identification number of "1". In the case of encoding angles, we allocate 16 bits of data. For instance, if a real steering angle α falls within the interval $k\Delta\alpha < \alpha \leq (k+1)\Delta\alpha$, α corresponds to the k th interval, and the binary value of k is used as the encoded value. During decoding,

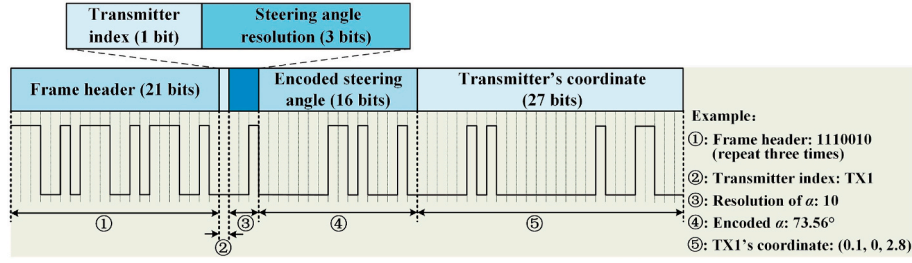


Fig. 3. OWPAE frame structure.

Table 3

Look-up table for OWPAE frame data.

Index of transmitter	
TX1	0
TX2	1
Resolution of steering angle	
9	000
10	001
11	010
12	011
13	100
14	101
15	110
16	111

upon receiving the binary value of k , it is converted to a decimal value, and α is calculated as $(k+0.5)180^\circ/2^{Nd}$. Additionally, we allocate 9 bits for each of the X, Y, and Z coordinates of the transmitter, respectively, to convey data. This frame structure enables us to apply the OWPAE scheme to compute the 3D coordinates and azimuthal angle of the receiver effectively.

4. Experiments and discussions

4.1. Experimental setup

In this section, we conduct experiments to evaluate the positioning and azimuth estimation performance of the proposed OWPAE scheme.

Fig. 4(a) shows the block diagram of our experimental setup. At the transmitter, we use Matlab running on a personal computer (PC) to generate optical signal waveforms based on the OWPAE frame structure. These waveforms are then loaded into an arbitrary waveform generator (AWG, Rigol DG2052). The AWG is responsible for driving two off-the-shelf line lasers to transmit data frames carrying the steering angle information in their respective directions. At the receiver, we employ a pair of PDs (Thorlabs, PDA100A2) to detect the line-shaped laser beams. After optical-to-electrical signal conversion, a digital storage oscilloscope (DSO, Rigol DS1000Z) records the waveforms, which are then transferred to a PC for offline processing. The offline decoding process includes synchronizing the frame header, extracting data bit information, and retrieving the steering angles. Using the proposed scheme, we can determine the 3D coordinate and azimuthal angle of the receiver. During the actual experiments, we manually adjust the steering angles of the two line lasers so that they alternately scan along the X- and Y- axes. Since the aperture of the PD is larger than the width of the received line-shaped beam, each PD can receive several data frames as the beams sweep through them. At each test location, we emit a test waveform from each line laser and scan it in close proximity to the PDs. By observing the DSO, we record the minimum and maximum steering angles that can be received by the PDs. Using these values of steering angles, we generate the corresponding data frames, encode them for retransmission by the line lasers, and capture them with the PDs. After reading from these data frames, the average value of the steering angles is calculated as the actual steering angle of each PD for 3D positioning and azimuth estimation.

Fig. 4(b) presents a photograph of the experimental setup at the

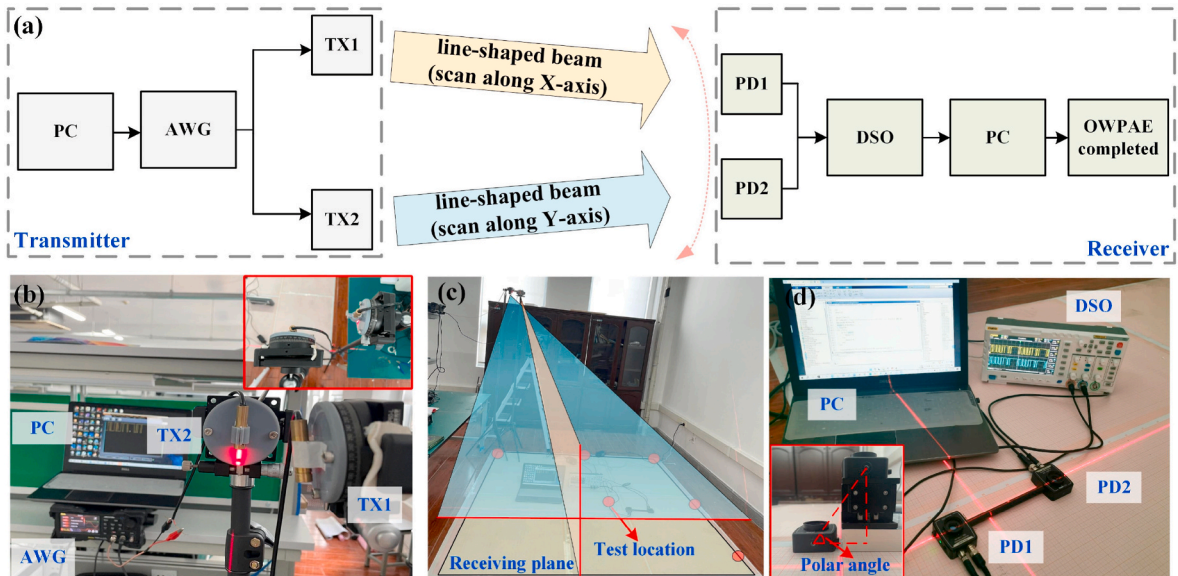


Fig. 4. (a) Block diagram of the experimental setup; (b) photograph on the transmitter side; (c) photograph of the complete experimental setup; (d) photograph on the receiver side.

transmitter, along with a top-down view subplot. The height-adjustable bracket holds two line lasers with a peak wavelength of approximately 650 nm. In our simulated indoor space, we consider these lasers to be mounted on the ceiling. The fixed positions of the line lasers within this coordinate system are (0, 0.1, 2.8) and (0.1, 0, 2.8) meters, respectively. For all experiments in this section, except for those in Fig. 9, we set the resolution of the steering angle at 12. Fig. 4(c) displays a photograph of the complete experimental setup. During the experiment, we simulate the scenario of positioning and azimuth estimation in an indoor space measuring $4\text{ m} \times 4\text{ m} \times 2.8\text{ m}$. The receiver is placed on the ground, with the ground serving as the receiving plane. We consider two different heights: 0.8 m and 1.6 m, corresponding to scenarios where the receiver is either placed on a table or held at chest level. To achieve this, we adjust the vertical distance between the line lasers and the ground to 2 m to simulate the receiver's position at a height of 0.8 m. For a receiver height of 1.6 m, we only need to adjust the height of the line lasers to 1.2 m. Considering the symmetry of the receiving plane, we select 6 representative test locations within 1/8 of the receiving plane to evaluate the OWPAE performance across the entire receiving plane. These locations correspond to the test locations in Fig. 2(a). When the height is 0.8 m, the coordinates of the test locations are (0, 0, 0.8), (1, 0, 0.8), (2, 0, 0.8), (1, 1, 0.8), (2, 1, 0.8), and (2, 2, 0.8) meters, respectively. For a height of 1.6 m, their Z-coordinate can be changed to 1.6. Fig. 4(d) presents a photograph of the experimental setup at the receiver. Except for Fig. 8, the distance between the two PDs is set at 0.1 m. The aperture of each PD is 9.8 mm, and its responsivity is 0.45 W/A when the wavelength is 650 nm. It is important to note that in our experiment, we primarily focus on two cases where the polar angle of the tilted receiver is 0° and 30° . When the polar angle is 0° , meaning the receiver faces upward, we simply fix the two PDs together using a support rod. When the polar angle is 30° , for proof-of-concept of our proposed scheme, we keep the two PDs facing upward to maintain the reception of LOS signals from the line lasers and adjust the horizontal and vertical distances of the two PDs according to the tilt angle and the distance between the PDs. The two PDs have a limited field-of-view (FOV). If each PD is tilted with a polar angle, e.g., 30° , the OWPAE frames emitted from the line lasers might be obstructed by the raised edge around the PD's aperture. Therefore, we position two PDs facing upward for proof-of-concept. Without FOV limitations, if the PDs are tilted along a rotational axis, the demodulated steering angle will remain unaffected because the PDs' locations do not change in our experiment. To ensure the reliable implementation of our scheme while accommodating the tilt of PDs, two approaches can be considered: using PDs with a larger FOV and deploying more line lasers to enhance transmit diversity. Specific placement details can be found in the subplot in Fig. 4(d).

It is worth mentioning that when using a laser-based positioning system, there is a trade-off in the integration advantage of positioning and illumination compared to traditional OWP systems. Nevertheless, the OWPAE system can not only coexist with lighting facilities but also operate independently, providing a more flexible and convenient deployment option that is conducive to fast deployment regardless of the lighting system's condition. On one hand, when deployed together, the original lighting system is responsible for providing illumination, while our system focuses on positioning. Our experiments are conducted under normal illuminance conditions ranging from 240 lux (at night) to 310 lux (during the day) at the location of the PD. These illuminance levels meet the requirements for various tasks, such as computer work (30 lux), conversation (100 lux), and reading (300 lux) [43]. Consequently, our scheme can coexist with lighting systems without interference. On the other hand, in scenarios where lighting systems are turned off to conserve energy, such as unmanned warehouses, traditional OWP systems may not function normally. In contrast, our proposed system can effectively address this issue, ensuring reliable positioning even when the lighting system is inactive.

5. Experimental results and discussions

First, we assess the positioning performance of our OWPAE scheme. We begin by examining the positioning accuracy under the influence of receiver tilt. Fig. 5 provides a top-down view of the experimental results for 3D positioning, illustrating 2D positioning outcomes at a height of 0.8 m for three distinct receiver tilt angles. In the figure, "+" represents real test locations, while "." represents the estimated results for these test locations. From the observations in Fig. 5, it is evident that our proposed scheme consistently achieves precise indoor positioning in the XY-plane, even when the receiver is tilted at various tilt angles. For instance, at polar angles of 0° , 30° , and 60° , the mean 2D positioning errors are 4.7 cm, 6.6 cm, and 9.4 cm, respectively. Notably, the positioning error experiences a slight increase as the receiver's polar angle increases. This can be attributed to the reduced distance between the PD projections on the XY plane as the polar angle increases. As a result, the difference between the values of the two steering angles (e.g., α_1 and α_2 or β_1 and β_2) becomes smaller, leading to a decreased denominator in Eq. (10). Consequently, even a minor quantification or measurement error can trigger more significant height estimation error, ultimately impacting positioning accuracy. Despite this reduced tolerance to positioning error due to receiver tilt, the proposed OWPAE scheme maintains its effectiveness, consistently achieving a 2D positioning error below 10 cm when the polar angle is less than 60° .

Next, we evaluate the positioning performance of the OWPAE scheme with the receiver placed at two different heights: 0.8 m and 1.6 m. At each height, we consider two cases with the receiver tilt angles set to 0° and 30° , respectively. Fig. 6 shows the experimental results for 3D positioning as the receiver is positioned at different heights and polar angles. These results illustrate that most of the estimated 3D locations closely align with their real locations, confirming the effectiveness of the proposed OWPAE scheme across different heights. In detail, when the polar angle is 0° , the mean 3D positioning error is 6.3 cm at a height of 0.8 m and 4.5 cm at a height of 1.6 m. When the polar angle is 30° , the mean 3D positioning error increase to 8.3 cm at a height of 0.8 m and 6.2 cm at a height of 1.6 m. By comparison, the 3D positioning accuracy is closely associated to the receiver height. This connection arises from the fact that higher-positioned PDs have a shorter distance from the line lasers, resulting in a more significant difference between the two steering angles, thus enlarging the denominator in Eq. (10). Consequently, quantization and measurement errors exert less impact on positioning accuracy. Therefore, greater receiver height provides the OWPAE system with improved tolerance to positioning errors.

Next, we conduct a statistical analysis of the system positioning performance under different receiver tilt angles. Fig. 7 displays the experimental results of cumulative distribution functions (CDFs) of 3D positioning errors at a height of 0.8 m. It is important to note that during experiments, quantization errors in steering angles and geometric measurement errors in the system can affect positioning accuracy. To provide a basis for comparison, we also present simulation results that assume ideal measurement readings of system parameters and no quantization errors in steering angles. From Fig. 7, it is clear that the OWPAE scheme can achieve a high probability of accurate positioning under varying tilt angles. For instance, the experimental curves show that 90 % of the 3D positioning errors are less than 12.8 cm and 14.3 cm, respectively, when the polar angle is 0° and 30° . Simultaneously, the simulated curves indicate that when the polar angle is 0° and 30° , 90 % of the 3D positioning errors are less than 5.1 cm and 12.9 cm, respectively. The simulation results consistently outperform the experimental results. This difference is due to the fact that simulation results are obtained under ideal conditions, without the impact of quantization and measurement errors encountered in the real-world scenario. Therefore, it is reasonable to consider the simulation results as an optimized upper limit for the experimental results.

As terminal devices often vary in size, and in our OWPAE scheme, PDs are mounted on these devices, we adjust the distance between the

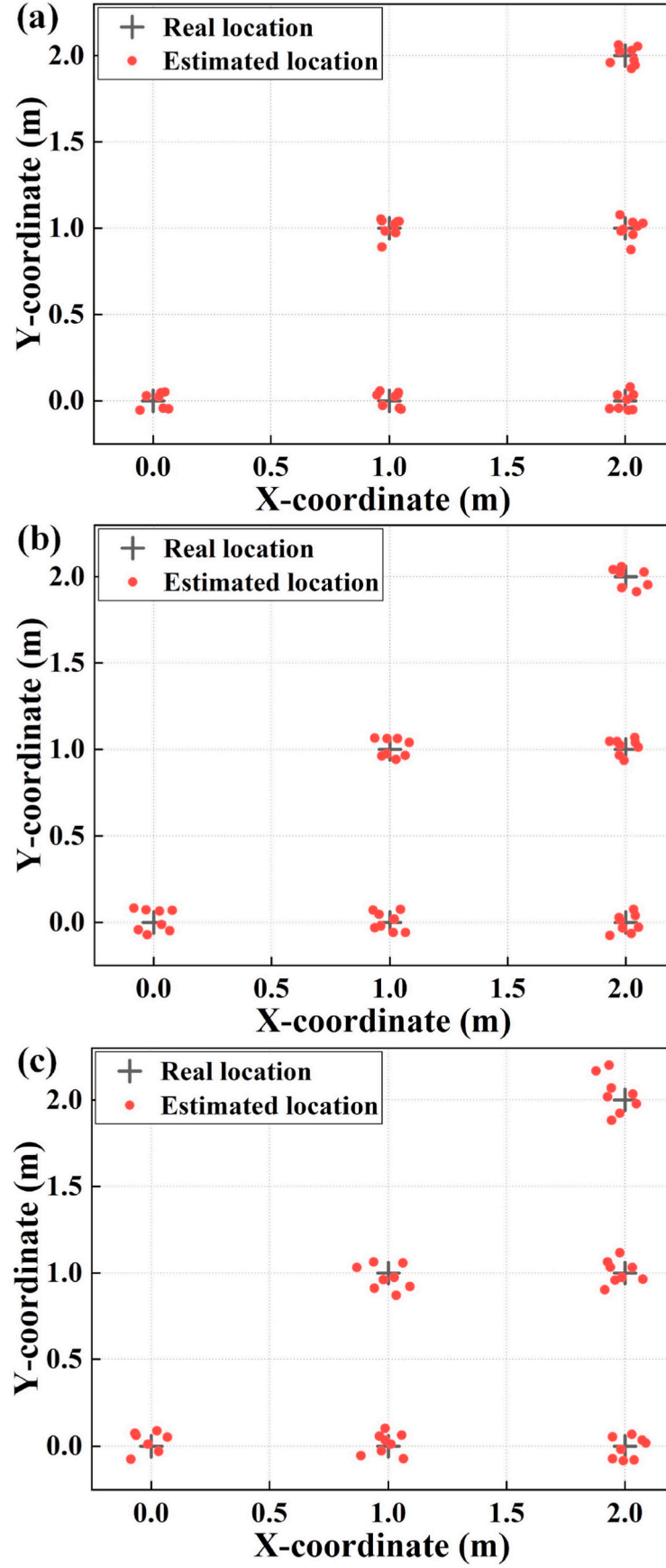


Fig. 5. Top-down view of the experimental results for 3D positioning at a height of 0.8 m for varying receiver tilt angles: (a) $\phi = 0^\circ$; (b) $\phi = 30^\circ$; and (c) $\phi = 60^\circ$.

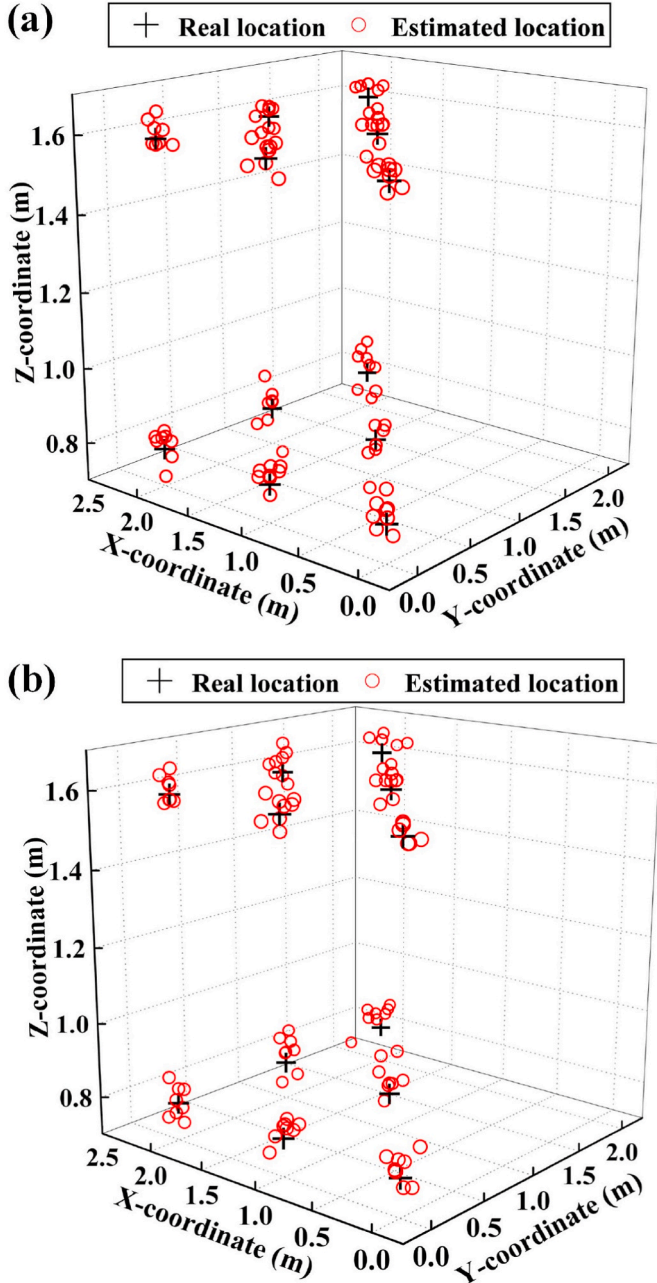


Fig. 6. Experimental results for 3D positioning at varying heights and tilt angles: (a) $\phi = 0^\circ$; and (b) $\phi = 30^\circ$.

PDs to assess the potential positioning and azimuth estimation performance of the OWPAE scheme. Fig. 8 illustrates the experimental results of mean 2D and 3D positioning errors, as well as azimuthal errors for the OWPAE system at a height of 0.8 m, under different PD distances and various receiver tilt angles. We considered three scenarios with PD distances of 0.1 m, 0.2 m, and 0.5 m, which simulates PDs mounted on a mobile phone, a robot vacuum cleaner, and a shopping cart, respectively. According to Fig. 8, the proposed OWPAE system consistently achieves high accuracy in positioning and azimuth estimation at different PD distances. Specifically, when the PD distances are 0.1 m, 0.2 m, and 0.5 m, and the polar angle is 0° , the mean 3D positioning errors are 6.3 cm, 4.8 cm, and 2.2 cm, and the mean azimuthal errors are 2.8° , 1.8° , and 1.2° , respectively. When the polar angle increases to 30° , the mean 3D positioning errors are 8.3 cm, 6.3 cm, and 4.7 cm, and the mean azimuthal errors are 5.6° , 2.9° , and 2.0° , respectively. It is evident

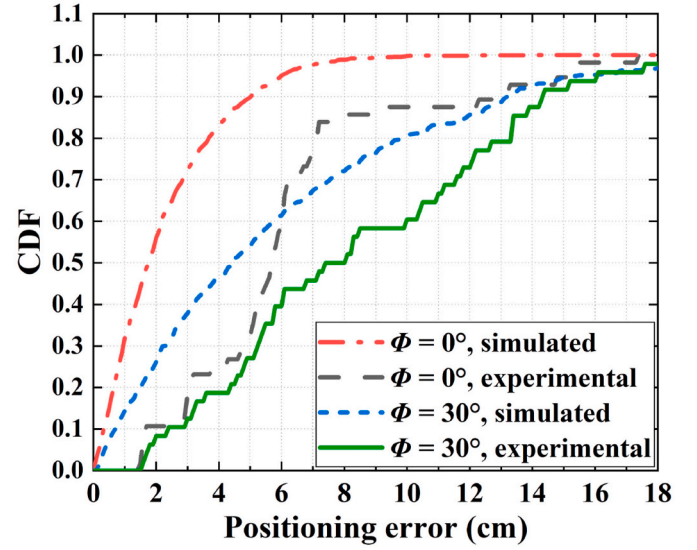


Fig. 7. CDFs of 3D positioning errors under different receiver tilt angles.

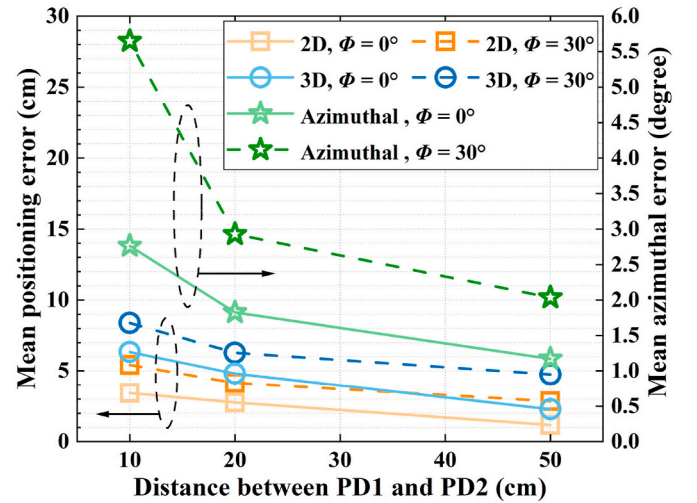


Fig. 8. Experimental results of mean positioning errors and azimuthal errors at different PD distances.

that as the distance between the PDs increases, both positioning and azimuth estimation accuracy improve. This phenomenon aligns with the analysis presented in Figs. 5 and 6, because a greater PD distance results in a larger difference between the two steering angles. In this context, quantization and measurement errors exert less influence on the overall positioning and azimuth estimation performance.

To evaluate the performance of the OWPAE scheme under different steering angle resolutions, we vary this resolution and observe its effect on positioning and azimuth estimation accuracy. Fig. 9 presents the experimental results of mean 2D and 3D positioning errors, along with azimuthal errors for the OWPAE scheme at a height of 0.8 m. The resolutions of the steering angles are set at 10, 11, and 12, representing angular increments of 0.1758° , 0.0879° , and 0.0439° , respectively. It is worth noting that in our experiment, the resolution of our steering device is 0.1° , indicating that the increment used to control the actual steering angles α or β of the line laser is 0.1° . When the line laser is directed towards a specific actual steering angle, the angular increment $\Delta\alpha$ or $\Delta\beta$ (the increment of the encoded steering angle emitted by the line laser) is determined by the steering angle resolution, i.e., $180^\circ/2^{Nd}$, rather than 0.1° . Therefore, the emitted encoded steering angle can be

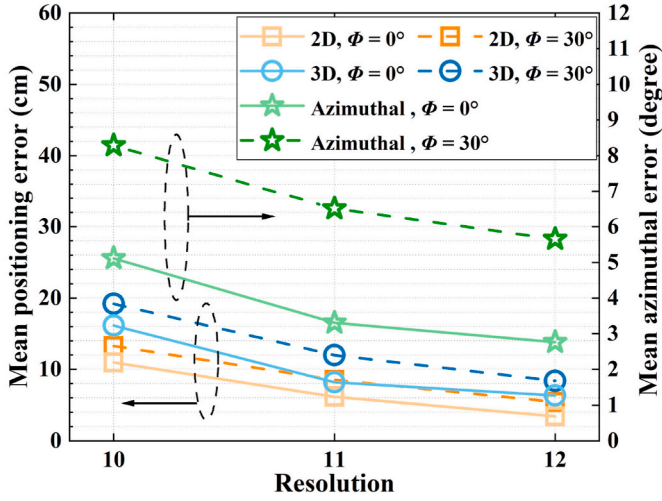


Fig. 9. Experimental results of mean positioning errors and azimuthal errors under different steering angle resolutions.

regarded as a quantization value of the actual steering angle of the line laser. With different N_d values of 11 or 12 adopted, where the angular increments less than 0.1° , different quantization values and quantization errors persist, leading to varied positioning performance. In theory, the quantization error with N_d of 12 should be smaller than that with N_d of 11. The results in Fig. 9 confirm that higher resolutions can enhance the positioning and azimuth estimation performance of the OWPAE scheme. When the resolution is set at 10, 11, and 12, and the receiver polar angle is 0° , the mean 3D positioning errors are 16.1 cm, 8.2 cm, and 6.3 cm, and the mean azimuthal errors are 5.1° , 3.3° , and 2.8° , respectively. When the polar angle is 30° , the mean 3D positioning errors are 19.2 cm, 11.9 cm, and 8.3 cm, and the azimuthal errors are 8.3° , 6.5° , and 5.6° , respectively. These results illustrate that improving the resolution, even if it is only in the transmitted angular information, can reduce the quantization error of the steering angle, thereby enhancing indoor positioning and azimuth estimation accuracy. This is because, with a higher steering angle resolution, the overall probability that the encoded steering angle is closer to α or β is higher. As a result, the mean positioning error obtained by a large number of measurements at different locations is lower. Given the current system configuration, for receivers tilted within the range of 0° to 30° , if positioning errors of around 20 cm and azimuthal errors within 10° are acceptable, a resolution of 10 is sufficient. However, if positioning errors need to be reduced to about 10 cm and azimuthal errors to within 6° , a resolution of 12 would be more suitable.

Next, we conduct a statistical analysis of the system azimuth estimation performance under different steering angle resolutions. Fig. 10 shows the experimental results of CDFs of azimuthal errors at a height of 0.8 m. For this analysis, we only consider the case when the receiver polar angle is 0° . In addition, we also present simulation results, as in Fig. 7, to provide a basis for ideal measurement readings of system parameters and no quantization errors in steering angles. From Fig. 10, it is evident that the OWPAE scheme can achieve a high probability of accurate azimuth estimation under varying steering angle resolutions. As the resolution increases, the azimuth estimation accuracy improves, which aligns with the conclusions in Fig. 9. For example, the experimental curves show that 90 % of the azimuthal errors are less than 8.1° , 5.8° , and 3.6° when the resolution is 10, 11, and 12, respectively. Simultaneously, the simulated curves, which can be considered as an optimized upper limit for the experimental results, indicate that when the resolution is 10, 11, and 12, 90 % of the azimuthal errors are less than 4.8° , 2.3° , and 1.0° , respectively.

Finally, we compare the proposed OWPAE scheme with Refs. [31, 32], both relying on steerable lasers. The comparison was conducted in

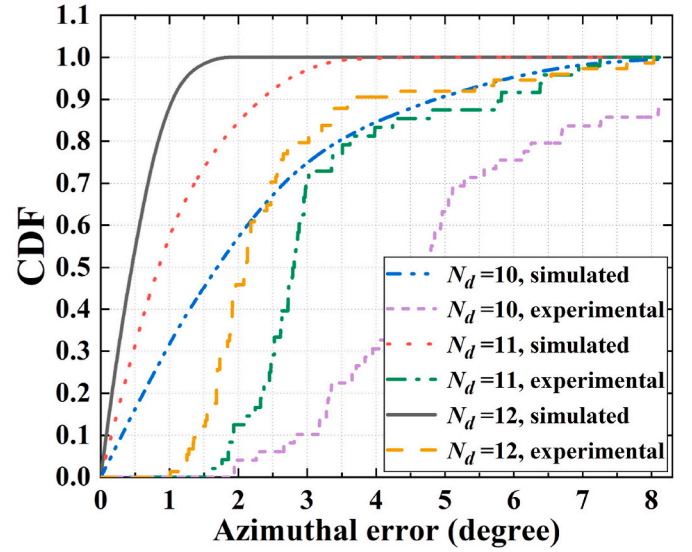


Fig. 10. CDFs of azimuthal errors under different steering angle resolutions.

three aspects. Firstly, in terms of the device, Ref. [31] require six steerable lasers and three 2D steering devices for 3D positioning, while in Ref. [32], four LEDs are required, and a laser is steered by a 2D steering device. Our scheme requires only two line lasers steered by two 1D steering devices, which is simpler and more cost-effective. Secondly, regarding complexity, [31] involves extensive calibration work to ensure the perpendicularity of lighthouses and measure various parameters crucial for positioning accuracy. [32] requires measuring or fitting the radiation model of the LED for positioning. In contrast, our scheme only requires accurate and perpendicular coordinates for the two line lasers, simplifying the setup and calibration process. Thirdly, considering functionality, [31] is designed for Smart Dust, supporting only positioning and experimentally verifying 2D positioning. [32] focuses on 3D positioning. In comparison, our scheme experimentally implements 3D positioning and azimuth estimation. In conclusion, the proposed OWPAE scheme exhibits superiority in terms of a simple system, low implementation complexity, and multifunctionality.

6. Conclusion

We proposed an OWPAE scheme using two steerable line lasers and a pair of PDs. This scheme can simultaneously estimate the 3D coordinates and azimuth of the receiver without a priori knowledge of radiation pattern of light sources and PD characteristics, irrespective of whether the receiver is tilted or not. Through experiments, we established a relationship model between the PD received voltage and the line laser driving voltage to verify the feasibility of using line lasers for OWPAE. We also validated the effectiveness of the proposed OWPAE in real-world scenarios. When the receiver is oriented upwards, the scheme exhibits accurate positioning performance with a mean 3D positioning error of 6.3 cm, where 90 % are less than 12.8 cm, and a mean azimuthal error of 2.8° . Even when the receiver is tilted at a polar angle of 30° , the scheme remains effective, yielding a mean 3D positioning error of 8.3 cm, where 90 % are less than 14.3 cm, and a mean azimuthal error of 5.6° . Results also indicate that the accuracy of the OWPAE system can be improved by increasing the receiver height, the distance between PDs, and the steering angle resolution. Quantization and measurement errors during the experiment significantly impact positioning and azimuth estimation accuracy. Effectively reducing these errors could bring the experimental results closer to their optimized upper limit.

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CRediT authorship contribution statement

Yuzhe Sun: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Xiaodi You:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Jian Chen:** Writing – review & editing, Methodology, Conceptualization. **Changyuan Yu:** Writing – review & editing, Resources, Methodology, Funding acquisition. **Mingyi Gao:** Writing – review & editing, Validation, Methodology, Investigation. **Gangxiang Shen:** Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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